GRAVITYSORT · PRODUCT REQUIREMENTS AND SPECIFICATIONS

GravitySort product requirements and specifications

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/03-requirements.md @ 6c7e5bc		

REVISION HISTORY

Version	Date	Author	Change
1.1	2026-10-25	Amish Chadha	Kit 1.8 pipeline: becomes GVS-PRS-001 (replaces GVS-REQ-001); R1 to R14 migrated to PR-001 to PR-014 with the old number in Was, each traced to a user requirement in GVS-URS-001, with tolerance and units; PR-015 (tipping with a full tank) and PR-016 (containment of a liner fragment) added; status from GVS-CAL-001 v0.9 after the STEM fixes F1 to F5 (GVS-DEC-001): PR-009 $594 (USD 139 over the target), PR-011 99.6 kg
1.0	2026-10-02	Amish Chadha	Status from GVS-CAL-001 v0.8 with the 1.7 m braced tank post and the flush container (decisions of 2026-10-02): R9 $580 (USD 125 over the target), R11 99.3 kg, R12 lowest safety factor 16, R13 hasps on both containers; no status changed
0.9	2026-10-02	Amish Chadha	

			R2 restated as 200 kg/h of feed time (GVS-DEC-001, 2026-10-02); status updated
0.8	2026-10-01	Amish Chadha	Status from GVS-CAL-001 v0.6 with the table bump stop (GVS-DDR-003 A2); R9 $569 (USD 114 over the target); R11 met on paper at 96.5 kg
0.7	2026-10-01	Amish Chadha	R11 restated by Amish (total 100 kg or less, loads 30 kg or less; GVS-DDR-003, A1); R11 met on paper
0.6	2026-10-01	Amish Chadha	Status from GVS-CAL-001 v0.4 for the constructable design (GVS-DDR-003); R11 not met (95.7 kg, every load under 30 kg); R9 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R9 target $450; R12 restated for the pedal drive; R11 met with the lighter frame tube; status from GVS-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 status from GVS-CAL-001; R7 pack range aligned with MotionCore and GVS-DDR-001 item 6; MotionCore cost $335; proposed changes to R11 and R12 listed, awaiting Amish
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

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What the design must do, each requirement traced to a user need in GVS-URS-001. Targets are proposals for review with a co-design partner, not user-validated needs. Method: STANDARDS section 22; the links to evidence are generated in GVS-RTM-001 (`03b-traceability.md`).

At TRL 3 every requirement has been checked by calculation or design review in GVS-CAL-001 v0.9, for the constructable design of GVS-DDR-003 with the STEM fixes of 2026-10-25 (GVS-DEC-001 F1 to F5). None is unmet. Four are at risk (PR-005, PR-006, PR-011, PR-016), two cannot be verified before testing (PR-004, PR-014), PR-009 is reported against the value-engineering target (USD 139 over it), and the other nine are met on paper or by design review. Recovery, liner life and containment can only be verified by testing, which is TRL 4 work and on hold under the portfolio cap.

Table 1. Product requirements and specifications. Status from GVS-CAL-001 (`docs/04-calcs/results.csv`).

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Safety	No mercury	The flowsheet shall have no amalgamation step, no mercury-coated plates or traps and no step that needs mercury to recover gold from the concentrate	0	none	steps using mercury	Design review of the flowsheet, TRL 3	Met (design review)	R1
PR-002	UR-002	Performance	Throughput	The machine shall process ore milled below 2 mm as slurry of 25 to 35 % solids at the design feed rate, giving about 1.55 t in an eight-hour shift with three flush stops (restated by Amish on 2026-10-02, GVS-DEC-001)	200	min	kg/h of feed time	Analysis, bowl capacity and cycle (GVS-CAL-001 section 3), TRL 3; timed run at TRL 4	Met on paper: 200 kg/h; 1.55 t per shift	R2
PR-003	UR-003, UR-014	Performance	Bowl G and speed shown	Bowl acceleration at the 100 mm reference radius shall be adjustable over the range, with the bowl speed shown to the operator	40 to 80	about 600 to 850 rpm	G	Analysis, drive ratio (GVS-CAL-001 section 2), TRL 3; inspection of the display at TRL 4	Met on paper: 40 to 81 G at a 50 to 71 rpm cadence; speed display (BOM line 19)	R3
PR-004	UR-003	Performance	Fine-gold recovery	Of free gold in the screened feed, the bowl concentrate	80	min (50 for 38 to 75 um;	%	Test with characterized	**Not verifiable at TRL 3; at risk.** Settling	R4

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				shall hold at least the target share of 75 to 1,000 um gold and 50 % of 38 to 75 um gold, and at least 60 % of all gold in the ore shall reach smelted gold		60 overall)		or spiked ore, TRL 4	check passes (STM S2); bed retention not calculable (STM S3)	
PR-005	UR-004	Performance	Smeltable concentrate	The bowl mass pull shall be at most 0.5 % of the feed, and the shaking table shall reduce one shift's bowl concentrate (about 5.3 kg) to the target mass	100	max (pull 0.5 % max)	g	Analysis, mass balance (GVS-CAL-001 section 8), TRL 3; test at TRL 4	**At risk:** pull 0.33 % met; 5.3 kg needs 53:1 on the table, likely two passes (STM S15)	R5
PR-006	UR-005	Performance	Pedal power	Full throughput shall need no more than the target power at the pedals, at a cadence of 55 to 70 rpm	60	max	W	Analysis, power estimate (GVS-CAL-001 section 7), TRL 3; measured at TRL 4	**At risk:** 48.2 W at 60 G met; 62.8 W at 80 G not; union seal drag assumed	R6
PR-007	UR-006	Electrical	Motor option	The MotionCore 250 W reference drive on any 20 to 58 V pack that MotionCore accepts shall run full throughput with at least the target power margin	3	min	times the power needed	Analysis (GVS-CAL-001 section 7), TRL 3	Met on paper: 5.2 times at 60 G, 4.0 times at 80 G; 0.55 kWh per shift	R7

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-008	UR-007	Performance	Water use and quality	Total water at 200 kg/h shall not exceed the target, and the fluidization supply shall tolerate recirculated water with fine silt (filtered to at most a third of the 1.0 mm hole size)	1.5	max	m3/h	Analysis, water balance and supply (GVS-CAL-001 sections 4 and 6), TRL 3	Met on paper: 1.19 m3/h; 120 mesh screen filter on the fluidization hose (GVS-DEC-001 F2); with the 3/4 in valve and rotameter (F6), 12.6 kPa at the union and 10.0 kPa net at the lowest ring at 600 rpm at mid-tank	R8
PR-009	UR-008	Physical architecture	Parts cost	Parts cost, excluding the MotionCore module and battery, shall be reported against the value-engineering target (a hypothetical control target, not a limit; Amish, 2026-10-01)	455	reported, not capped	USD	Analysis, priced BOM (GVS-CAL-001 section 12), TRL 3	Estimated cost $594: **over the value-engineering target by USD 139**	R9
PR-010	UR-009	Materials	Local workshop build	Every part shall be made with welding, drilling and hand tools, with no lathe; the bowl liner cast in a printed mold	0	none	parts needing a lathe	Design review of every part (GVS-BLD-001), TRL 3; build at TRL 4	Met (design review); casting route unproven (STM S25)	R10
PR-011	UR-010	Packaging and transport	Transport mass	The machine shall break into loads carried by two	100	max (each	kg	Analysis, mass estimate (GVS-	**At risk:** 99.6 kg in six loads, heaviest	R11

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				people and weigh no more than the target in all (restated by Amish on 2026-10-01, GVS-DDR-003 A1), assembled with hand tools in 30 min or less		load 30 max)		CAL-001 section 11), TRL 3; weighed at TRL 4	27.3 kg, only 0.4 kg of margin on an estimate (independent review); assembly time not verified	
PR-012	UR-011	Safety	Guards, speed and stopping	The bowl shall run only under its lid guard; belts, chains and the table head shall be guarded; with the motor, bowl speed shall be capped at 900 rpm by the ratio and the MotionCore limit; with the pedals, the bowl, jacket and rings shall keep a burst safety factor of at least 10 at 1,200 rpm with the speed shown; the bowl shall stop within 15 s of stopping the drive (restated per GVS-DDR-002)	10	min (stop 15 s max; motor 900 rpm max)	safety factor	Design review, burst calculation (GVS-CAL-001 section 10), TRL 3	Met on paper: guards, lid interlock and disc brake (0.6 s stop); motor capped; lowest safety factor 16 at 1,200 rpm	R12
PR-013	UR-012	Usability		The bowl concentrate	5	max	min	Design review, TRL	Met on paper:	R13

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
			Quick, secure clean-up	shall be flushed into a lockable container without tools within the target time, and the table concentrate shall drop into a lockable box				3; timed trial at TRL 4	toolless lid clamps; flush container and concentrate box with padlock hasps (GVS-DEC-001); time not verified	
PR-014	UR-013	Materials	Liner life and replacement	The bowl liner and riffles shall be replaceable in 30 min, with a liner life of at least the target	500	min (replace in 30 min max)	h	Wear data for cast polyurethane on quartz, TRL 4	Not verifiable at TRL 3 (STM S29)	R14
PR-015	UR-011	Safety	Stability with a full tank	The machine with a full header tank shall stand on a slope of at least the target in any direction without tipping (criterion decided by Amish on 2026-10-03, GVS-DEC-001), and the tank post and braces shall not yield at the side push that tips it	10	min	degrees	Analysis, tipping and post check (GVS-CAL-001 section 11), TRL 3	Met on paper: 13.1 degrees with the 560 mm drum (GVS-DEC-001 F1); post safety factor 2.1 at a 185 N push; sway of the full tank near the table drive frequency to decide (GVS-DEC-001 O3)	none
PR-016	UR-011	Safety	Containment of a liner fragment	The tub and lid shall contain a quarter of the liner and shell thrown off at	1200	min	rpm	Analysis (GVS-CAL-001 section 10), TRL 3; impact test	**At risk:** contained face first (57 J against 40 J); not shown edge	none

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				the highest reachable speed (decided by Amish on 2026-10-01: check by calculation before any spin test; steel band only if the check fails)				before any spin test, TRL 4	first (14 J against 22 J at 900 rpm) or for a quarter of the whole bowl group (125 J); band option awaiting Amish (GVS-DEC-001 O2)	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- Ore is milled to below 2 mm by the user's existing mill and screened on the GravitySort hopper screen.
- Free, liberated gold in the 38 to 1,000 um range. Gold locked in sulfides or finer than about 20 um is lost to tailings by any gravity method, so PR-004 applies to free gold only.
- Reference ore for the daily balance: 5 g/t, 1.55 t per shift (7.75 g of gold). Real grades vary widely (about 1 to 20 g/t).
- Healthy adult pedalling at 60 W or less for sessions of about 1 h, with operators taking turns.
- Budget: the $455 in `project.yaml` is a value-engineering target, a hypothetical control target and not a limit (Amish, 2026-10-01); it was raised from $350 to $450 on 2026-09-25 and to $455 on 2026-09-26 (GVS-DDR-002). It covers GravitySort parts. The MotionCore kit and reference motor ($335: $265 kit plus $70 motor, per MTC-CAL-001) and a battery are shared lab components, budgeted with MotionCore.
- The pack for the motor option is left to the user (GVS-DDR-001 item 6).

Changes to requirements decided by Amish

- 2026-09-25 (GVS-DDR-002): R9 target $350 to $450; R12 restated for the pedal drive (burst safety factor of 10 or more at 1,200 rpm plus the speed display); R11 target kept at 80 kg with the lighter frame tube.
- 2026-09-26: R9 target $450 to $455 (budget approved).
- 2026-10-01 (GVS-DDR-003, A1): R11 restated from 80 kg or less to 100 kg or less, every load 30 kg or less.
- 2026-10-02 (GVS-DEC-001): R2 restated as 200 kg/h of feed time, about 1.55 t per eight-hour shift with three flush stops.
- 2026-10-03 (GVS-DEC-001): the tipping criterion behind PR-015, a 10 degree slope in any direction with a full tank.

Changes on 2026-10-25 (kit 1.8 pipeline; not decisions by Amish)

- The R numbers became PR numbers, traced to GVS-URS-001; no target changed.
- PR-015 and PR-016 written down from decisions Amish had already made (2026-10-03 and 2026-10-01).
- PR-008 now names water quality for the fluidization holes, because the STEM check found that the design had no filter (STM S24; filter added under the standing instruction of 2026-10-23, GVS-DEC-001 F2).